



BUSINESS REQUIREMENTS DOCUMENT
AI-Powered Contract Intelligence & Clause Extraction Platform

Project-wise Contract Repository with Automated Parsing, Clause Extraction, Contract Snapshot, AI Insights & Semantic Search

Domain	Category	Priority	Status
Legal & Secretarial	ROI / Productivity Enhancement	High	Draft

Version	Prepared For	Prepared By	Date
1.1	Hackathon Evaluation	Business Analyst	01 Aug 2026

SOLUTION SUMMARY

The platform converts unstructured legal agreements into actionable business intelligence by combining document parsing, LLM-based classification, clause extraction, semantic search, risk identification, and AI-generated insights.

1. Introduction

1.1 Purpose

This Business Requirements Document defines the business need, scope, functional requirements, technical approach, database structure, user journey, dashboard expectations and acceptance criteria for C.L.E.A.R. — an AI-powered Contract Intelligence Platform.

The solution will help business and legal users transform unstructured agreements into a centralized, searchable and insight-driven repository. It will classify documents, extract key clauses, identify risks, surface important dates and provide concise contract summaries through dashboards.

WHAT THIS MEANS

Instead of reading each agreement manually, users can view a structured contract snapshot and drill into clause-level evidence with page references.

1.2 Background

Organizations often store agreements across shared drives, email folders and document repositories. These contracts may be in PDF, scanned PDF or Word format, and the key terms are generally embedded in unstructured legal text.

- Manual review consumes significant legal and operational effort.
- Important obligations such as renewal, expiry, payment terms and liability may be difficult to track.
- Search results are inconsistent when documents are not indexed by clause, party, date or contract type.
- A standardized AI-assisted process can improve speed, traceability and governance.

2. Business Objectives

- Centralize contract documents in a structured repository.
- Classify agreements into predefined document types, such as NDA, MSA, SLA, Contract Cum Order Form and Licensing Agreement.
- Extract important clauses and summarize them in business-readable language.
- Detect missing, non-standard, and high-risk clauses.
- Provide page-level traceability for every extracted clause.
- Enable keyword and semantic search across agreements.
- Generate contract snapshots and AI insights for faster business decision-making.
- Support Legal, Procurement, Finance, Compliance and Business stakeholders.

3. Scope

3.1 In Scope

Area	Included Capabilities
Document Management	Project-wise repository, single/bulk document ingestion, document metadata, versioning and status tracking.
Supported Documents	NDA, MSA, SLA, vendor agreements, client contracts, licensing agreements, lease agreements and other legal agreements.
Supported Formats	PDF, scanned PDF, DOC and DOCX.
AI Processing	OCR, parsing, JSON generation, document type classification, clause extraction, semantic chunking, embeddings and summarization.
Search & Analytics	Keyword search, clause search, semantic search, repository search and dashboard insights.
Reporting	Contract summary export, clause summary export, risk reports and contract inventory reports.

3.2 Out of Scope

- Contract drafting and authoring.
- Contract negotiation workflow and redlining.
- Digital signature or e-signature capability.
- External CLM integration in Phase 1.
- Legal advice or final legal interpretation.
- Multi-language clause extraction unless approved as a later enhancement.

4. Stakeholders

Stakeholder	Primary Responsibility
Legal Head	Business sponsor, governance owner and final approval authority for clause interpretation framework.
Legal Team	Reviews AI output, validates extracted clauses, corrects mappings, and approves summaries.
Procurement Team	Searches vendor agreements for tenure, payment terms, termination and renewal details.
Finance Team	Reviews contract value, payment obligations, liability and closure-related terms.
Business Teams	Submit contracts, review snapshots and consume business insights.
Compliance Team	Monitors risk indicators, missing clauses and contract governance exceptions.
System Administrator	Manages projects, users, roles and Clause Master configuration.
Solution Team	Builds and maintains document processing, AI pipeline, database, search and dashboards.
Business Analyst	Owns requirements, process documentation and stakeholder validation.

5. Current State and Future State

5.1 As-Is Process

1. Contracts are stored as individual files.
2. Users manually open documents to locate relevant clauses.
3. Information is copied into spreadsheets, emails or notes.
4. Renewal and expiry obligations are tracked outside the contract repository.
5. Review outcomes depend on manual interpretation and available time.

5.2 Key Challenges

- High manual effort and delayed response times.
- Limited visibility into contract obligations.
- Inconsistent search and retrieval.
- Risk exposure due to missed clauses or dates.
- No centralized contract intelligence layer.

5.3 To-Be Process

The future workflow moves the organization from scattered, manually-tracked files to a single automated pipeline. Each contract passes through six stages, from the moment a user logs in to the moment results are searched and exported; Section 6 walks through this same flow step by step with full detail.

Stage	What Happens
User Access	The user logs in with organizational credentials. Role-based access determines which projects, documents and actions they can see and perform.
Project Selection	The user chooses the project or repository category the contract belongs to, so it is filed and governed under the right business context.
Contract Submit	The user uploads one document or a batch. The system checks format, file size, duplicate status and access rights before accepting it.
AI Processing	The backend runs OCR/parsing, document-type classification, clause extraction and embedding generation automatically, with no manual data entry.
Snapshot & Insights	The system produces a business-friendly Contract Snapshot and AI Insights covering key terms, important dates, and risk or missing-clause flags.
Search & Export	The user finds contracts and clauses through keyword or semantic search, then exports summaries, clause data or risk reports as needed.

The future process will automate document understanding and provide structured contract information through a repository, dashboards, semantic search and role-based review workflow.

6. End-to-End User Journey

Step	Process	Description
1	Access Platform	User logs in via Single Sign-On (SSO) using organizational credentials. Role-based access determines which projects, documents and actions are available.
2	Select Project	User selects the relevant project or repository category where the contract should be stored.
3	Submit Contract	User submits one or more documents. The system validates format, size, duplicate status, access rights and project mapping.
4	AI Processing	The backend reads the document, extracts structure, identifies document type, detects clauses, creates embeddings and generates summaries.
5	Contract Snapshot	User views a business-friendly summary with key terms, important dates, risk indicators and document status.
6	Clause Analysis	User reviews extracted clauses with summaries, page references, risk level, source text and confidence score.
7	Legal Validation	Legal reviewer accepts, corrects, comments on or overrides AI-generated output before final review completion.

Step	Process	Description
8	Risk Monitoring	Users track missing clauses, high-risk language, non-standard terms, upcoming renewals and auto-renewal obligations.
9	Search Repository	Users search by clause, counterparty, date, document type, value, keyword or natural language query.
10	Export Results	Users export summaries, search results, risk reports and dashboard data for reporting or review.

7. AI Processing Architecture

7.1 Backend Flow

Stage	Brief Description
Document	Source legal agreement available for backend processing.
ADI Processing	OCR and parsing engine extracts text, layout, tables, headings and page references.
JSON Output	Structured representation of title, section headings, tables, paragraphs and page numbers.
LLM Document Type Identification	LLM identifies the contract type based on title, sections, key phrases and legal context.
Section Heading Check	System checks whether meaningful section headers or titles are available.
Header-Based Clause Detection	If headings exist, LLM identifies clauses using document type and section context.
4-Page Chunking	If headings do not exist, content is split into four-page chunks for semantic clause discovery.
Clause Mapping	Identified clauses are mapped to the Clause Master and tagged with confidence and risk indicators.
Embedding Generation	Clause text and chunks are converted into vectors to support semantic search and AI retrieval.
Data Persistence	Document-level metadata is stored in Document Master and clause content is stored in Document Content Master.
Dashboard & Search	Contract Snapshot, AI Insights, searchable repository and reports are generated.

PROCESSING PRINCIPLE

The backend flow begins with Document, not Document Upload. Upload is a user action, while Document is the starting point of internal AI processing.

8. Data Model

8.1 Document Master

Document Master is the document-level database table. It stores the core identity, location, classification, processing and review status for each contract.

Column	Description
DocId	Unique system-generated document identifier.
ProjectId	Project or repository category linked to the document.
DocumentName	Original uploaded file name.
DocumentPath	Storage path of the source document.
JsonPath	Storage path of the parsed JSON output.
DocumentType	Classified contract type such as NDA, MSA, SLA or Vendor Agreement.
CounterpartyName	Vendor, client or other contracting party extracted from the document.
ContractValue	Monetary value extracted from the document, where available.
EffectiveDate	Contract start date.
ExpiryDate	Contract end date.
ContractTerm	Duration or tenure of the agreement.
ProcessingStatus	Current AI processing status, such as Pending, In Progress, Processed or Failed.
ReviewStatus	Business review status, such as Pending Review, Reviewed or Exception.
UploadedBy / UploadedDate	User and timestamp for document submission.
CreatedOn / ModifiedOn	System timestamps for record creation and updates.

8.2 Document Content Master

Document Content Master stores the clause-level knowledge extracted from each document. It supports semantic search, clause dashboards, evidence traceability and AI retrieval.

Column	Description
ContentId	Unique clause or content record identifier.

Column	Description
DocId	Foreign key linking content to Document Master.
ClauseName	Extracted clause name, such as Termination, Liability or Payment Terms.
ClauseType	Clause category used for reporting and risk analysis.
SectionHeader	Section or title from which the clause was extracted, where available.
ReferencePage	Page number where the clause appears in the contract.
Content	Extracted clause text.
ChunkId	Identifier of the four-page chunk if chunk-based processing was used.
Embedding	Vector representation used for semantic search.
ClauseSummary	AI-generated explanation of the clause.
RiskLevel	Risk indicator, such as Low, Medium or High.
MissingClauseFlag	Indicates whether a mandatory clause was not found.
ConfidenceScore	AI confidence score for extraction accuracy.

8.3 Clause Master

Clause Master defines what the system should identify, validate and monitor. Admins can use it to configure mandatory clauses, keywords, definitions, risk scores and document-type applicability.

Clause Master Field	Description
ClauseId	Unique clause identifier.
ClauseName	Clause or key term title.
ApplicableDocType	Document type where the clause is expected or relevant.
Keywords	Keywords, patterns or phrases used for detection.
Description	Business/legal definition of the clause.
MandatoryFlag	Defines whether the clause is mandatory for a document type.
RiskWeight	Risk score used to calculate clause or contract risk.
ActiveFlag	Indicates whether the clause is currently active for extraction.
VersionNo	Configuration version used for traceability.

9. Dashboard Requirements

9.1 Contract Snapshot

PURPOSE

The Contract Snapshot gives users a quick business-friendly view of a contract without reading the full document.

Snapshot Area	Data Displayed
Contract Information	Document Type, Counterparty, Contract Value, Status and Project.
Key Terms	Payment Terms, Governing Law, Auto Renewal, Notice Period and Liability Cap.
Important Dates	Effective Date, Expiry Date, Renewal Date, Notice Date and Contract Duration.
Clause Overview	Total clauses found, missing clauses, mandatory clauses and high-risk clauses.
Summary	Concise AI-generated explanation of the agreement and its major obligations.

9.2 AI Insights Dashboard

Insight Category	Description
Risk Insights	Highlights high-risk clauses, unusual language, missing protections and liability exposure.
Renewal Insights	Highlights upcoming renewals, auto-renewal terms and expiring agreements.
Business Insights	Summarizes payment obligations, commercial commitments, vendor dependencies and key legal obligations.
Review Insights	Identifies contracts requiring legal review, low-confidence extractions and clauses needing validation.

10. Functional Requirements

Module	Requirement Summary
Document Processing	Support PDF, scanned PDF, DOC and DOCX. Extract text and structure using ADI/OCR and generate structured JSON.
Document Classification	Automatically identify document type using LLM.
Clause Extraction	Extract clauses using section headers where available and semantic chunking where headings are unavailable.
Clause Mapping	Map extracted clauses to Clause Master and capture page-level references.
Risk & Missing Clause Detection	Identify missing mandatory clauses, high-risk terms and non-standard language.
Search	Support keyword, semantic, clause, counterparty and date-based search.

Module	Requirement Summary
Dashboard	Provide Contract Snapshot, Risk Dashboard, Renewal Dashboard and AI Insights Dashboard.
Administration	Support user roles, project management, Clause Master maintenance and configuration management.
Export	Allow export of summaries, extracted clauses, risk reports and search results.

11. Non-Functional Requirements

Category	Requirement
Security	Role-based access control, Single Sign-On (SSO) based authentication, secure storage, encryption and audit logging.
Performance	Efficient processing of large agreements and fast repository search response.
Reliability	Traceable extraction results with page references and recovery from processing failures.
Scalability	Ability to support growing volumes of documents, users, projects and search requests.
Auditability	Maintain detailed history of uploads, extraction results, reviews and overrides.
Usability	Simple, business-friendly interface for legal and non-legal users.
Compliance	Follow organizational data privacy, retention and access control policies.

12. Acceptance Criteria

ID	Acceptance Criteria
AC-01	Users can submit supported contract documents under a selected project.
AC-02	System validates document format, duplicate status, size and access permission.
AC-03	ADI/OCR extracts document text, layout, headings, tables and page references.
AC-04	LLM identifies the correct document type.
AC-05	Clauses are detected using section headers when available.
AC-06	Four-page chunking is used when section headers are unavailable.
AC-07	Extracted clauses are stored in Document Content Master.
AC-08	Document metadata is stored in Document Master.
AC-09	Each extracted clause displays source content and page number.
AC-10	Contract Snapshot displays type, value, auto renewal, governing law, key dates and term.
AC-11	High-risk and missing clauses are clearly highlighted.

ID	Acceptance Criteria
AC-12	Users can search contracts by clause, counterparty, date, document type, value or keyword.
AC-13	Users can export summaries, extracted clauses and reports.
AC-14	Legal reviewers can validate, update or override AI output.

13. Risks and Mitigation

Risk	Mitigation
Poor scanned document quality may reduce OCR accuracy.	Flag low-confidence text extraction and route for manual review.
Clause language may vary across legacy agreements.	Use Clause Master, LLM-based semantic matching and embeddings to improve detection.
Missing section headers may affect direct extraction.	Use four-page chunking and semantic clause discovery.
Document type may be misclassified.	Allow Legal Reviewer or Admin to override document type.
Sensitive contract data may be exposed.	Apply role-based access, secure storage, encryption and audit logs.
Clause Master may evolve during implementation.	Start with minimum mandatory clauses and allow version-controlled updates.

14. Expected Business Benefits

Benefit Area	Expected Impact
Operational Efficiency	Reduces repetitive manual review effort and accelerates contract analysis.
Searchability	Makes contracts searchable by clause, party, date, document type and business terms.
Risk Visibility	Highlights missing clauses, high-risk language and upcoming obligations.
Governance	Standardizes review output and improves traceability through page-level references.
Decision Support	Provides contract snapshots and AI insights for faster legal and business decisions.

15. Future Enhancements / Roadmap (Phase 2)

The items below extend capabilities already validated in Section 12 (Acceptance Criteria) and are proposed for a subsequent phase, subject to stakeholder prioritization and effort estimation.

15.1 Enhanced Clause Traceability (builds on AC-09)

Enhancement	Description
Inline Source Highlighting	In addition to the page-number reference (AC-09), highlight the exact source sentence/paragraph within an embedded document viewer so users can verify context without leaving the platform.
Side-by-Side Evidence View	Show the extracted clause text alongside the original document page image for faster validation by Legal Reviewers.
Multi-Clause Cross Reference	Link related clauses within a single contract (e.g., Termination referenced within Liability) so users can trace dependent obligations.

15.2 Advanced Search & Discovery (builds on AC-12)

Enhancement	Description
Natural Language Search	Allow users to type a plain-language question (e.g., “Which vendor contracts renew in the next 60 days?”) in addition to structured search by clause, counterparty, date, document type, value or keyword (AC-12).
Value & Range-Based Search	Extend keyword/clause/counterparty/date search to support contract value thresholds and ranges.
Saved Searches & Alerts	Allow users to save frequent search criteria and receive a notification when new or updated contracts match.
Cross-Contract Semantic Search	Expand semantic search to identify similar clauses or risk patterns across multiple contracts and projects simultaneously.

15.3 Contract Assignment & Ownership (New Capability)

WHAT THIS MEANS

Beyond finding and reading contracts, users will be able to assign a contract or a review task to a specific person or team, so ownership, due dates and accountability are tracked directly in the platform.

Enhancement	Description
Contract Assignment	Allow Admin/Legal Head to assign a specific contract (or a contract review task) to a named user or team for ownership, action and accountability.
Assignment Rules	Support assignment by Project, Document Type or Counterparty, with manual reassignment where required.
Assignment Notifications	Notify the assignee when a contract is assigned, reassigned, or nearing its review/renewal due date.
Ownership Dashboard	Provide a dashboard view of contracts grouped by assignee, showing status, due dates and overdue items.
SLA & Escalation	Track assignment SLAs and escalate to the Legal Head if a review is not completed within the agreed timeframe.

15.4 Future Acceptance Criteria (Indicative)

ID	Acceptance Criteria
FAC-01	In addition to the page number (AC-09), users can view the highlighted source sentence/paragraph for each extracted clause within an embedded document viewer.
FAC-02	In addition to clause, counterparty, date, document type, value and keyword search (AC-12), users can search contracts using natural-language queries.
FAC-03	Users can save search criteria and receive alerts when new or updated contracts match.
FAC-04	Admin/Legal Head can assign a contract or a contract review task to a specific user or team.
FAC-05	Assignees receive a notification when a contract is assigned, reassigned, or approaching its due date.
FAC-06	A dashboard view shows contracts grouped by assignee with status and due-date tracking.

16. Glossary

Term	Definition
ADI	Azure Document Intelligence or equivalent document parsing engine used to extract text, layout and structure.
LLM	Large Language Model used for classification, clause identification, summarization and insights.
Document Master	Database table containing document-level metadata, classification and processing status.
Document Content Master	Database table containing clause-level extracted content, page references, embeddings and summaries.
Clause Master	Admin-managed list of clauses and key terms used as extraction criteria.
Embedding	Vector representation of text used for semantic search and retrieval.
Contract Snapshot	Dashboard view summarizing important contract information.
AI Insights	AI-generated observations related to risk, obligations, missing clauses and review actions.
Chunking	Splitting content into smaller sections, such as four-page chunks, for better clause identification.

FINAL SUMMARY

C.L.E.A.R. converts unstructured contracts into a structured intelligence repository through ADI parsing, LLM classification, clause extraction, semantic search, Contract Snapshot dashboards and AI-generated risk insights.